

Aaron Ware

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SENIOR APPLIED AI ENGINEER | BACKEND & PLATFORM SYSTEMS

Applied AI and backend/platform engineer building production systems for LLM-guided workflows, rules-driven application engines, payments, and complex business operations. Strongest at turning high-stakes domain logic into reusable infrastructure: MCP servers, agentic interfaces, headless workflow engines, streaming APIs, and cloud-deployed services.

SKILLS

AI & Agentic Systems: Model Context Protocol, LLM tool use, RAG, prompt architecture, Anthropic SDK, OpenAI API, Voyage embeddings, streaming LLM interfaces

Backend & Data: Node.js, TypeScript, Express, REST APIs, MongoDB, PostgreSQL, DynamoDB, Supabase, Stripe, clean architecture, dependency injection

Frontend: React, Next.js, SvelteKit, SolidJS, Chakra UI, TailwindCSS, CSS Modules, Vite

Cloud, Testing & Tools: AWS ECS Fargate, Cognito, Secrets Manager, ALB, SST, Pulumi, GitHub Actions, Vercel, Render, Jest, React Testing Library, Mocha, Chai

EXPERIENCE

Senior Applied AI Engineer / Backend Lead

Buddy Technology | Richmond, VA | Jun 2021 - Present

Insurance technology startup building embedded distribution, payments, and AI-guided application infrastructure.

- Architected Buddy's production Model Context Protocol server in TypeScript and Node.js, enabling LLM clients to guide users through end-to-end workflows including field collection, quoting, payment, document Q&A, and policy binding.
- Designed per-session MCP transports backed by Buddy's application engine, managing field visibility, validation, view progression, disqualification logic, and asynchronous third-party quote flows across complex multi-step applications.
- Managed development of a Next.js/React agentic chat application while leading prompting strategy, including model selection, prompt caching, tool-use rules, and optimization of multi-turn workflows.
- Architected and shipped ITX, a platform-agnostic TypeScript framework that runs Buddy's internal product-modeling DSL across web apps, Stripe Apps, MCP servers, AI agents, and future platform targets.
- Decoupled the core engine from React, creating a shared state manager, action engine, and UI orchestration layer that powers complex workflows across web, Stripe Apps, MCP servers, and headless consumers.
- Built RAG-powered document Q&A using MongoDB Atlas vector search, Voyage embeddings, and score thresholding to support semantic search over policy and product documents.
- Designed payments infrastructure using dependency injection, factory, and adapter patterns; implemented ACH payment microservices for asynchronous processing, observability, and logging, contributing to 2x monthly written premium and sustained 100% YoY growth.
- Led backend implementation for Buddy's core transaction and policy domains, including payments, refunds, ledgers, analytics events, partner integrations, and policy lifecycle workflows.

- Created and maintain a repository of internal agent skills that aid engineering teamwork through domain-specific context, guided repository onboarding, and repeatable development workflows.

Digital Engagement & Product Manager

Elevation Advertising | Richmond, VA | May 2019 - Jun 2021

- Led cross-functional discovery and delivery for client websites and digital products, translating business goals into requirements, user flows, scopes of work, and implementation priorities.
- Defined, prototyped, and shipped an MVP proof-of-concept application using Google Sheets, Apps Script, JavaScript, and PostgreSQL to validate workflow assumptions before deeper engineering investment.
- Conducted user research, stakeholder interviews, competitive analysis, and analytics review to identify customer needs, prioritize product opportunities, and inform roadmap recommendations.

SELECTED PROJECTS

LactSafe

SvelteKit, TailwindCSS, DaisyUI, Serverless, OpenAI API

app.lactsafe.com

Application that makes LactMed data easier for lactating people to retrieve and understand through accessible search and AI-assisted summarization, answering approximately 120 medicine safety questions per month.

Folio

Astro, Claude Code, AI-assisted content generation, developer tooling

folio.earnwhere.dev

Claude Code-powered portfolio and engineering journal generator that interviews users, personalizes site scaffolding, drafts first-person about and project content, and generates engineering journal posts from coding sessions.

MyHotelReports

SvelteKit, TypeScript, Express, Supabase, PostgreSQL, Puppeteer PDF generation, Render

www.myhotelreports.com

Hospitality reporting platform for operators and property owners, supporting month-end statement workflows, configurable PDF reports, owner messaging, reservation visibility, and backend-gated access to financial and operational data.

EDUCATION

Bachelor of Science, Small Business Management and Ideation

Virginia Commonwealth University | 2015

Invited back after graduation to serve as an adjunct professor and Entrepreneur-in-Residence.